

HARISH REDDY A

 harishreddya11@gmail.com  +91 6360899660  Bengaluru, India

 [linkedin.com/in/harishreddya8](https://www.linkedin.com/in/harishreddya8)  github.com/harishreddya11

PROFESSIONAL SUMMARY

Generative AI and Machine Learning engineer with hands-on experience building and deploying NLP applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, embeddings, and vector databases. Strong Python skills with a solid foundation in linear algebra, probability, and statistics, plus practical experience in data pre-processing, feature engineering, model evaluation, and deploying AI models to production using FastAPI and Docker.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, HTML5, CSS3
- **Generative AI & NLP:** Large Language Models (LLMs), GPT, OpenAI API, Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Embeddings, Semantic Search, Text Pre-processing, Tokenization, Speech Recognition, Speech-to-Text, OCR
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Data Pre-processing, Feature Engineering, Model Evaluation, Exploratory Data Analysis
- **Python Libraries:** Pandas, NumPy, Matplotlib, BeautifulSoup, Requests, SQLAlchemy
- **Mathematics:** Linear Algebra, Probability, Statistics, Discrete Mathematics
- **Vector & Databases:** pgvector, PostgreSQL, MySQL
- **Backend & Deployment:** FastAPI, REST API Design, JWT Authentication, Alembic, Docker, Swagger/OpenAPI, Postman, Git, GitHub
- **Computer Science:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, SDLC, Debugging, Problem Solving

PROJECTS

DevPilot AI – Generative AI Developer Assistant (RAG-Based NLP Application)

Python, FastAPI, LangChain, OpenAI API, pgvector, PostgreSQL, React, Docker | github.com/harishreddya11/DevPilot-AI

- Built an end-to-end generative AI application answering developer questions over private documents using Retrieval-Augmented Generation (RAG) with LangChain and OpenAI GPT models.
- Implemented the full NLP pipeline: OCR document ingestion, text cleaning and chunking, embedding generation, and semantic search over a pgvector vector database.
- Tuned chunk size, overlap, and top-k retrieval parameters experimentally, evaluating retrieval quality on a labelled question set to reduce irrelevant context.
- Deployed and maintained the application in a containerized production environment using Docker, exposing the model via secure, versioned FastAPI REST endpoints with JWT authentication, applying prompt engineering and context-window management to reduce hallucinated responses.
- Designed a modular backend using SQLAlchemy ORM, Alembic migrations, and the Repository-Service pattern, documented via Swagger/OpenAPI.

Voice Assistant – Speech-to-Text NLP Application

Python, Speech Recognition, Natural Language Processing | github.com/harishreddya11/Voice-Assistance1

- Developed a voice-controlled assistant that converts spoken audio to text and maps user utterances to executable commands using natural language processing techniques.
- Built intent-matching logic over normalized, tokenized text to classify user commands into action categories.
- Improved command recognition accuracy through iterative error analysis on misrecognized inputs and refined text normalization rules, with code managed under Git.

EXPERIENCE

Data Science Intern — Skill Forge E-Learning Solutions Pvt. Ltd., Bengaluru

Jun 2024 – Jul 2024

- Processed and analysed 10,000+ records in Python using Pandas and NumPy, performing data cleaning, preprocessing, and exploratory analysis to surface operational trends.
- Automated repetitive data preprocessing and feature preparation workflows, reducing manual effort by approximately 30%.
- Built Python web scraping pipelines to collect and structure external datasets for downstream modelling and analysis.
- Wrote clean, documented code and technical documentation to improve reproducibility across the team.

EDUCATION

Bachelor of Computer Applications (BCA) — Bengaluru North University

2021 – 2024

CGPA: 9.36/10 | Coursework: Probability & Statistics, Discrete Mathematics, DBMS, Object-Oriented Programming

CERTIFICATIONS

Software Engineer (HackerRank) | Python (HCL) | Google Cloud Digital Leader | AWS Solutions Architecture Job Simulation (Forage)